Yujian Zhang

(+86) 15620773019 | yjzhang0720@gmail.com

Education

Tianjin University - **Master of Computer Science and Technology** Sep 2022 - Jun 2025

Rank **1/22** (87.41/100) Conducted research in semi-supervised learning and domain generalization, published a related conference paper, and received multiple scholarships and programming competition awards.

Sichuan University - **Bachelor of Water Science and Engineering (Outstanding Graduate)** Sep 2018 - Jun 2022

Rank **1/58** (86.17/100) Applied computational analysis to water purification research, published several papers, and received the National Scholarship and the Outstanding Graduate of Sichuan Province honor.

Research

Research on Semi-Supervsed Domain Generalization - **Student 1st author** Sep 2022 - Sep 2023

Multi-level Augmentation Consistency Learning and Sample Selection for Semi-Supervised Domain Generalization

Accpeted by 2024 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2024)

- Improved generalization with limited labeled data. The study combined image- and feature-level style augmentation with explicit constraints on domain-invariant features, helping the model perform on test data from different distributions.
- Made training more efficient. It introduced a strategy for selecting high-quality samples to label and achieved state-of-the-art results on the PACS and OfficeHome domain generalization benchmarks.

Cross Research Papers of Machine Learning and Water Purification - **1st author/Co-author** Sep 2019 - Jul 2025

Data analysis and knowledge mining of water treatment experimental results using classical machine learning methods. For example, we use linear regression to predict the concentration of novel pollutants in water. The related research has published 9 SCI papers, 3 papers of first author, and currently I have **more than 650 citations** on Google Scholar.

Work Experience

TikTok - **Algorithm Engineer** Jan 2025 - Present

- Develop and optimize recommendation algorithms for TikTok Shop livestreams, with a focus on improving conversion efficiency and the viewer experience.
- Improved recommendation model architecture, substantially increasing conversion efficiency without significantly increasing costs.
- Balanced livestream commerce goals with the broader TikTok experience, improving conversion outcomes while continuing to enhance users' short-form video experience.

Baidu - **Algorithm Intern** Jun 2024 - Jan 2025

- Fine-tuned video-language models for anomaly detection and improved traffic detection across cities, increasing mAP50 by 0.7%–11.0% on real-world data.
- Built a vehicle classification model from scratch with 95.7% average accuracy and delivered six computer vision projects.

Lenovo - **Algorithm Intern** Mar 2024 - Jun 2024

- Evaluated multimodal models for document localization in complex scenes using a new dataset, and explored lightweight segmentation approaches.
- Reproduced leading semantic and instance segmentation models and refined their outputs using multiscale image context and post-processing.

Publications

Yu M, **Zhang Y**, Li X, et al. Multi-Level Augmentation Consistency Learning and Sample Selection for Semi-Supervised Domain Generalization[C]//ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2024: 6550-6554. (**Student 1st author**)

Zhang Y, Zhou G, Yue J, et al. Enhanced removal of polyethylene terephthalate microplastics through polyaluminum chloride coagulation with three typical coagulant aids[J]. Science of the Total Environment, 2021, 800: 149589. (**1st author**)

Zhang Y, He L, Wang G, et al. Defluorination and regeneration study of lanthanum-doped sewage sludge-based activated carbon[J]. Journal of Environmental Chemical Engineering, 2021, 9(4): 105740. (**1st author**)

Li Z, **Zhang Y**, Wei X, et al. A comprehensive study on the performance and mechanism of microplastics removal by enhanced coagulation methods[J]. Journal of Water Process Engineering, 2023, 56: 104238. (**Equal 1st author**)

Xing X, **Zhang Y**, Zhou G, et al. Mechanisms of polystyrene nanoplastics adsorption onto activated carbon modified by ZnCl2[J]. Science of The Total Environment, 2023, 876: 162763. (2nd author)

Huang L, He W, **Zhang Y**, et al. Chitosan enhances poly aluminum chloride flocculation system removal of microplastics: effective, stable, and pollution free[J]. Journal of Water Process Engineering, 2023, 54: 103929. (3rd author)

**Only selected publications are listed.*

Honors & Awards

- National Scholarship
- National Encouragement Scholarship
- Outstanding Graduate of Sichuan Province
- Outstanding Student, Sichuan University
- Special Prize Scholarship for Graduate Students
- Third Prize, National English Competition for College Students
- Third Prize, National University Student Social Practice and Science Contest on Energy Saving and Emission Reduction
- Gold Award, National College Students' Algorithm Design and Programming Challenge
- First Prize (Graduate Division), National College Students' High-Tech Competition — Algorithm Star Challenge

Skills

- Deep learning/machine learning frameworks: Numpy, PyTorch, scikit-learn, etc.
- Programming Language: Python, C++, Linux Shell.
- Others: Office, Origin, Adobe Premiere Pro, Cinema 4D.